

PK ENTITY LINKING ANNOTATION GUIDELINES FOR TABLES

Project Background

Biomedical Literature contains a wealth of information on Pharmacokinetics (PK), including parameters and covariates, often reported in semi-structured tables. This data is helpful for preclinical PK predictions and initiating Population PK Models. However, it takes time for PK researchers to manually extract information from the literature. Extracting this data automatically from free text would enable accelerated curation of PK databases for research and drug development.

Automatically extracting data from tables is a complex task. The machine needs to (a) recognise the entities within the table cells (e.g., PK parameters, units, numeric values), known as named entity recognition (NER) (b) be able to standardise these entities by linking different surface forms to a single form in a structured knowledge base or ontology, known as entity linking (EL) and (c) understand the relations between entities in cells of the tables.

Aim: In this study, we aim to develop an automated entity-linking approach to link the PK parameter mentions in table cells to concepts in the PK ontology. This addresses the entity linking step (b) of automated data extraction from tables in PK.

Method: We sample tables from PubMed Central Open Access papers reporting PK parameters. Then, we ask annotators to link these PK parameters as concepts from the PK Parameter ontology.

Task Overview & Annotation Interface

The annotation interface displays the text from a single cell of the table, with a single PK parameter mention highlighted. The full table and associated table caption and footer (if present) from which the cell originates are also presented, with the cell highlighted to give context. The interface also displays a drop-down list of selectable options representing the complete PK ontology. The annotator must select a label from these possible choices for the mention. It also shows a list of selectable options representing the concepts in the PK parameter ontology, including a “NIL” option for irrelevant (e.g. not a PK parameter in the ontology). Two annotators are assigned to

annotate each table; a third annotator checks disagreements and raises these to discuss with the team where necessary.

The main subtasks are as follows:

1. First, Check if the initial PK entity span at the top is correct; you can re-highlight it if there are any errors.
2. Secondly, choose the PK entity subtype from the drop-down list. The “NIL” label is for entities not associated with a value in the list.
3. Please note that the interface will not let you proceed to the following example if you do not choose an option from the drop-down list. Therefore, if you are not sure, please select the closest option or “NIL”, and flag the example and leave a comment.

Details of the key functionality of the labelling interface are shown below:

Table Data:

	0	1	2	3
0	0			
1	Tmax (h)	0.39 ± 0.04	k01 t1/2 (h)	0.12 ± 0.04
2	Cmax (ng/mL)	0.29 ± 0.07	k10 t1/2 (h)	0.67 ± 0.20
3	k01 (per h)	6.27 ± 1.57	AUC0-∞ (h ng/mL)	0.40 ± 0.12
4	k10 (per h)	1.10 ± 0.31	Vd/F	12.49 ± 2.38
5	k10 (per h)	1.10 ± 0.31	Cl/F	224.84 ± 59.89

FOOTER: Data represent five different horses. Tmax, time to maximum concentration; Cmax, maximum concentration; k01, first-order absorption rate constant; k10, first-order elimination rate constant; k01 t1/2, half-life of absorption; k10 t1/2, half-life of elimination; AUC0-∞, area under the concentration-time curve extrapolated to infinity; Vd/F, apparent volume of distribution dependent on bioavailability; Cl/F, clearance dependent on bioavailability.

Annotations:

- Save progress:** Floppy disk icon in the top left-hand corner.
- Span can be adjusted:** PK 1 label above the table.
- Flag example for review:** Flag icon in the top right-hand corner.
- Keep track of progress:** Progress bar on the left sidebar.
- Go back to up to 10 previous examples:** History list on the left sidebar.
- Drop down menu with suggestions:** absor| dropdown menu.
- Leave a comment:** Comment box below the dropdown.
- Accept Answer:** Green checkmark button.
- Reject Answer:** Red X button.
- Go back to the last example:** Left arrow button.
- Link to origin paper:** Link icon in the bottom right.

Saving Annotations: If your annotations are saved correctly, you will see a green box in the right-hand corner when you press the floppy disk (in the top left-hand corner). If this box is red, an error occurs on the server, so please raise this immediately.

Viewing Guidelines: You can quickly view these guidelines inside the interface by pressing the question mark in the top left-hand corner.

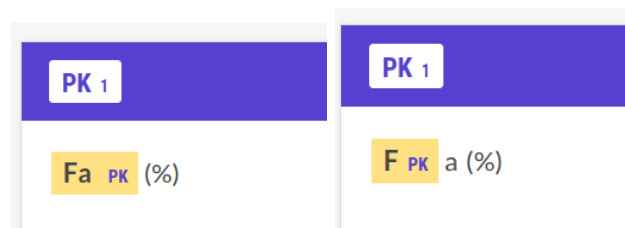
Further Information: A link to the original paper from which the table is taken is provided to allow information to be checked against the paper text if needed.

What if I'm unsure? Please flag the example using the flag icon in the top right-hand corner and leave a comment in the box. The team can then review and discuss it.

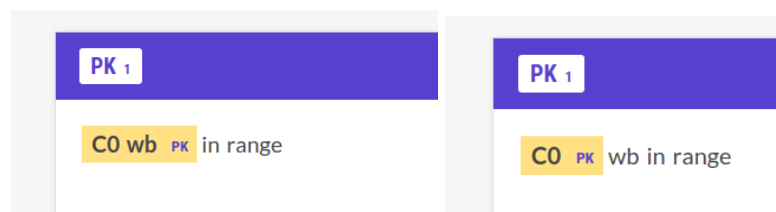
Summary of Doubt/Clarifications

Dealing with NER model errors:

- **Non-PK Parameters:** If the highlighted span refers to any of the following, please label them as “NIL”:
 - Part of a PK/PD parameter is highlighted (e.g. AUC/MIC).
 - A pure PD parameter is highlighted (e.g., MIC, Emax, Maximum tolerated dose (MTD), or time to disease progression).
 - Any other text that does not refer to in vivo PK parameters, e.g., physiological parameters, chemical parameters, or clinical endpoints.
- **Incorrect PK Spans:** If the highlighted span refers to a PK parameter, but the span is not correct, please correct the spans, for example:
 - If only part of the full span has been highlighted (e.g., if the model highlighted “Cl” of the full term “Clearance”), then please correct this to highlight the full span.
 - If the relative NER model incorrectly includes a footnote in the model name, please remove these, e.g.



- If the parameter highlighted includes extra text at the beginning or end of the span which is not part of the parameter itself, such as in the example below, please shorten the span:



- However, if a composite entity refers to a specialised version of a PK parameter, e.g. AUC_{sample}, Cmax{drug}, then link the entire span to the core PK parameter in the ontology.
- In cases such as variability on parameters (e.g. IIV Cl) or parameter covariate effects (e.g. weight on Cl), only link the parameter mention itself (e.g. Cl) to the ontology.
- Please note if the highlighted span does not refer to a PK parameter, please **do not** adjust/correct it, just label as “NIL”.

Other Hints and Tips:

- If a PK parameter is highlighted but is not included in our ontology, please include it but mark it with the “NIL” label.
- In cases where the stated parameter is similar to our labels or a combination of two or more of our labels, but not one specific type, we will mark it as NIL, for example:
 - “Cmax(ss)” combines C_{ss} and C_{max} from our ontology. We don’t have this combined parameter as an option, so link to NIL.
 - “t_{1/2} pi” or “t_{1/2}n” refers to very fast distribution phase before t_{1/2}alpha. - This is not included in our ontology, so it is classified as NIL.
- “Oral bioavailability” should be marked as absolute bioavailability (“F_{ab}”).
- Look out for the “TBC”, which can stand for Total Body Clearance.
- “AUC(0-tau)” - mark as AUC_{0-t} considering tau as a t, given we do not know tau from the parameter alone.
- Cumulative excretion or urinary recovery rates are marked as “Q_{44:fe}” if the units are % or a fraction. If the units are in mass, mark them as “Q_{91:Ae}”.
- Relative Bioavailability relative from one condition to another instead of F_{rel} - Please label this as NIL (a covariate effect and out of scope).